

Rupali Patel

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GitHub

LinkedIn

GeeksforGeeks

LeetCode

EDUCATION

Vellore Institute of Technology

B.Tech in Electronics and Communication Engineering

Cumulative GPA: 9.22/10

Bhopal, Madhya Pradesh

Expected June 2027

TECHNICAL SKILLS

Industrial & Embedded Systems: Sensors, Sensor Interfacing, Microcontrollers, Embedded C, Arduino

Communication Protocols: I2C, UART, SPI, Bluetooth

Programming: C, C++, Java

Engineering Tools: Arduino IDE, Proteus, LTspice

Technical Areas: IoT, Signal Processing, Hardware Debugging, System Integration

EXPERIENCE

Embedded System Design Intern

Jan 2025 – Mar 2025

Maven Silicon

- Developed embedded firmware and hardware interfacing for a women's safety alert system using microcontroller-based design.
- Integrated sensors and peripheral devices for reliable hardware-software communication.
- Strengthened practical knowledge of embedded programming, hardware debugging, and system integration.

PROJECTS

Pulse Oximeter

Jul 2024 – Sep 2024

- Designed and developed a real-time pulse oximeter using Arduino and MAX30100 to measure heart rate and SpO₂.
- Processed physiological signals using Embedded C and the I2C communication protocol.
- Displayed real-time measurements on an LCD interface.
- **Technologies:** Arduino, Embedded C, MAX30100, LCD, I2C

Object Detection in Water

Sep 2024 – Dec 2024

- Designed the complete hardware circuit and integrated ultrasonic sensors with Arduino.
- Developed an underwater object detection system capable of sensing submerged objects.
- **Technologies:** Arduino IDE, Embedded C, Ultrasonic Sensor, Serial Communication

Sign Language to Speech Conversion System

Jan 2025 – Apr 2025

- Integrated flex sensors and HC-05 Bluetooth with Arduino for real-time gesture acquisition.
- Developed a system that converts sign language into text and speech.
- **Technologies:** Arduino IDE, C++, HC-05 Bluetooth, Flex Sensors

CERTIFICATIONS

- Applied Machine Learning in Python
- MATLAB Signal Processing Onramp
- NPTEL Internet of Things

ACHIEVEMENTS

- Recipient of the Dream STAR Scholarship at VIT Bhopal University.
- Recipient of the Reliance Foundation Undergraduate Scholarship.
- Secured First Position in District Merit (Class XII).

ADDITIONAL INFORMATION

- **Languages:** English, Hindi
- **Interests:** Sketching, Reading, Gaming